

A note to the reader of Part I

The sequel proposed at the end of this paper is now written.

Part I (*Random Signs into Matchings*) closed with a proposal and a promise: to prove the **Heilmann–Lieb theorem** not through interlacing machinery but through **Godsil’s path tree**, and—*“if this paper has a sequel, this is its first sentence.”* That sequel now exists.

Part II — Unfolding a Graph into a Tree: A Machine-Checked Proof of the Heilmann–Lieb Theorem in Lean 4 carries out the proposal in full. It builds the path tree, proves the divisibility $\mu_G \mid \mu_{T(G,u)}$ (crossing the one elaboration wall the path-tree route had stalled at), establishes the forest identity, and proves the Ramanujan bound from a weighted Gershgorin estimate. Both halves are **sorry-free**.

For every finite graph G , the matching polynomial μ_G is real-rooted; and if $2 \leq \Delta$ and every vertex has degree $\leq \Delta$, every root of μ_G lies in

$$[-2\sqrt{\Delta-1}, 2\sqrt{\Delta-1}].$$

This is exactly the endpoint that Part I mapped as future work (“The next stone”): the first machine-checked Heilmann–Lieb, completing the path-tree program. As in Part I, the mathematics is classical and no new theorem is claimed; the contribution is the formalization.

Where to find Part II.

- Sources and both PDFs (English + Spanish): <https://github.com/karlesmarin/godsil-gutman-lean>
- Archived on Zenodo: DOI 10.5281/zenodo.20561832 (<https://doi.org/10.5281/zenodo.20561832>).